

Javier Robles

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Summary

I'm a full stack software engineer, proficient in Python and JavaScript. Creative problem solver with a strong passion for independent learning, collaboration, and creating unique human-centered interfaces. Currently pursuing an M.Eng. in Software Engineering at the University of Maryland. Background in Mechatronics engineering and manufacturing.

Technical Skills

Python: Django, Django REST API, Flask, SQLAlchemy, AWS API, Pillow, SQL ORM

JavaScript: React, React-Router, Typescript, Material UI, Motion, Plotly, Leaflet, Webpack

CSS: Tailwind CSS, Framer, Animations, Bootstrap, Transitions, Flexbox, Grid, Responsiveness

Databases: SQLite, PostgreSQL, MongoDB, Google Firebase

Amazon Web Services (AWS): S3, CloudFront, email services (SES)

Version Control: GitHub

Freelance Full Stack Experience

Lavandería Coco: www.LavanderiaCoco.com

Dec 2024 - Feb 2025

Developed a full stack web application for a local laundry service. The website administrator can manage clients, orders, line items and cash flow. Customers can keep track of the progress of their orders via WhatsApp notifications.

- Created a Django back end, with user authentication, and Django API SQL database ORM optimized for aggregations and quick querying.
- Implemented a clean UI, optimizing space to ensure responsive functionality, front end features components built with React.js, React Router, Tailwind CSS, google icons and Plotly.
- Managed the full development lifecycle, from initial design through deployment.

Etiqueta Sana: www.EtiquetaSana.com

Jul 2023 - Oct 2024

Web application and content management system for nutritionists, enabling collaboration on custom meal plans, client-specific recipes, a live food database, progress tracking, and blog management.

- Engineered the back end using Flask, SQLAlchemy ORM, Flask-Login, integrating with AWS S3 for secure storage of user uploaded media, and email template handling for application notifications.
- Implemented front end features with JavaScript, Bootstrap, Plotly and pure CSS, ensuring a responsive and user-friendly interface.
- Managed the full development lifecycle, from initial design through deployment.

Previous Experience

Design Engineer | Carrier HVAC Mexico

2019 - 2022

- Responsible for engineering change design review. Produced monthly reports on engineering change reworks.
- Managed bulk data updates impacting thousands of BOMs within a relational database.
- Normalized tables containing unique purchase parts into relational tables, resulting in a 90% reduction in total rows. This streamlined the part purchasing process.

Volunteer Experience

Survey Results Visualization | American Institute in Taiwan Spring 2023

- Analyzed survey responses from more than 300 employees regarding perceptions to identify variations between subpopulations.
- Created data visualizations in multiple languages for a diverse audience.

Projects

NYC LEP Speakers | javrobs.pythonanywhere.com/nyc_lep_speakers/

Sunburst and map visualization of Limited English Proficiency (LEP) residents of NYC.

- SQL database loaded with Pandas ETL. Django ORM querying and API. JS Plotly and Leaflet front end.

Austin Housing | www.javrobs.github.io/house_pricing_analysis/

Map visualization of house sale prices in Austin from 2018 to 2021, fed into a linear regression and neural network models for price estimation.

- Pandas ETL, TensorFlow and Scikit Learn models, Flask API, Leaflet and Bootstrap front end.

Education

M.Eng. in Software Engineering | University of Maryland

Sep 2025 – May 2027 (In progress)

- Coursework: AGILE development, Machine Learning, Secure coding practices.

Certificate of Data Analytics and Visualization | Tecnológico de Monterrey

Jan – Jun 2023

B.Eng. in Mechatronics | UANL

2014 - 2019

- French Mexican Agreement for Engineering Formation (MEXFITEC) scholarship holder.
- One year engineering program at the "Institut National des Sciences Appliquées" (INSA) in Lyon, France.